2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.311 Distributed Systems

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a. Describe what RPC is. Identify and describe three of the functionalities RPC provides to the system developer.

[5 marks]

1.b The lowest RPC guarantee is where there is no contingency if anything goes wrong. Discuss the three other delivery semantics, highlighting the associated trade offs.

[6 marks]

1.c Consider the following protocol implementation.

```
int send_message(recipient_address, content) {
  max_retrys = 4;
  ack_received = false;
  c = open_channel(recipient_address);
  for (i=0; i<max_retrys && ack_received==false; i++) {
    send(c, content); // push message content through the
network
    if( received(c, ack) ) { // non-blocking listening
      ack_received = true;
    }
  }
  return ack_received;
}
```

i. Describe the level of guarantee of this protocol. Explain what sort of messages/method calls this is suitable for.

[4 marks]

ii. What needs to be changed in order to provide a higher level of guarantee? You should explain which level that is and any assumption(s) you make about other parts of the implementation.

[4 marks]

1.d Explain what RMI offers that the RPC model does not.

[4 marks]

1.e In order for RMI to work, besides the client and server code which implement the method call and implementation logic, an additional entity is required. Describe this entity and its function.

[2 marks]

Total 25 marks

Question 2

2.a Describe what is meant by the term 'middleware'. Give an example to support your answer. **[4 marks]**

2.b Indirect communication is a fundamental concept in middleware systems to overcome a range of problems such as partial failure caused by failed nodes or variable connectivity. Describe what is meant by indirect communication and explain the two dimensions of uncoupling it seeks.

[6 marks]

2.c Message Queues are an example of an indirect communication approach.

i. State and compare the different ways in which messages can be retrieved. **[3 marks]**

ii. Explain one shortcoming of Message Queues. **[3 marks]**

2.d Explain with examples any two of the subscription models of publish/subscribe.

[6 marks]

2.e Describe one indirect communication abstraction that does not require the developer to deal with send/receive or put/get primitives. Describe how this abstraction works.

[3 marks]

Total 25 marks

Question 3

3.a Describe three functions that must be provided by a group communication service.

[3 marks]

3.b Consider the following pseudo code that has a group communication behaviour. The `onReceive` method (lines 11 – 13) will be automatically invoked when a message arrives. The `send` method (line 7) provides no guarantee of message delivery.

```
1   public class ToyProgram {
2
3    public static void main (int id) {
4       join("comm. group"); //Join the communication group
5       if (id > 0) {
6          for each member in the group {
7                 send("Hello "+ id);
8          }
9       }
10
11     void onReceive(Message msg) {
12          print(msg); //Print the received message.
13     }
14  }
```

i.

What type of message reception mode does this code implement? (1 mark)

What is the advantage of this reception mode over the alternative? (2 marks)

[3 marks]

ii. Assuming six (6) instances of this program are launched with ids -2, -1, 0, 1, 2, and 3 in this order, **answer the following questions and justify your answer**:

What is the maximum number of "Hello ..." messages to be printed out? (3 marks)

What is the minimum number of "Hello ..." message to be printed out? (2 marks)

[5 marks]

Question 3 continues on next page...

Question 3 continued…

3.c

i. Describe the three levels of reliability that a multicast protocol may provide (1.5 marks), and indicate precisely which guarantees each level provides (1.5 marks).

[3 marks]

ii. A local bank wants to develop a fault tolerant system using active replication. Their system only needs to support crediting operations (i.e. additions). They plan to use a group communication protocol to disseminate incoming requests across two servers, and do not envisage needing to dynamically deploy new servers.

Do they need any ordering guarantees to ensure both replicas behave consistently? (1 mark)

Why? (2 marks)

[3 marks]

3.d

i. Query flooding is a search mechanism. Explain how a newly-joining node publishes the files they wish to share in (1) a centralised p2p network (*2 marks*) and (2) a query flooding p2p network (2 *marks*).

[4 marks]

ii. Bitcoin was introduced in the p2p systems lecture. The Bitcoin protocol automatically adjusts the difficulty of the *proof-of-work* so that it takes around *10 minutes* before the first Bitcoin server solves the proof-of-work to produce a new block to be added to the existing block chain. The new block chain will then be broadcast to the Bitcoin p2p network so that other Bitcoin servers can stop their current proof-of-work and start working on the new block chain. Given the scale of the Bitcoin network, the new block chain takes more than *one minute* to reach the majority of the nodes in the network.

Ten minutes could be a long time for some users to wait for confirmation of their transactions, so somebody proposes a new protocol that only requires ***10 seconds*** to solve the proof-of-work.

Explain why this new protocol is a bad idea.

[4 marks]

Total 25 marks

Question 4

4.a Define the notions of *fault*, *error*, and *failure.* **[3 marks]**

4.b The Google Chubby lock service uses passive replication. Describe what passive replication is.

[2 marks]

4.c. Consider the following pseudo code that describes the core function of a Chubby-like system. The function will be invoked when a user request is received.

```
1 process(request) {
2     check & process request
3     send acknowledgement to client
4     for each of the 3 remaining replicas {
5         send update
6     }
7 }
```

In the following questions we assume that duplicated requests are properly handled by the system, and that there is a mechanism to safely switch to a new primary server if the current primary crashes**. We also assume that *only one client* can connect to the system at a time.**

i.
What type of replicas will this code be running on? (*2 mark*)
Present a *failure free* execution of the above function on a time-sequence diagram (3 marks).

[5 marks]

ii. One of the students in SCC.311 suggests replacing lines 4-5 with an *atomic multicast* operation. This new implementation is described below.

```
1 process (request) {
2     check & process reques
3     send acknowledgement to client
4     atomic_multicast(remaining_replicas[], update);
5 }
```

Do you think this modification is necessary? (1 mark)
Illustrate your explanations with an example using a time-sequence diagram. (3 marks)

[4 marks]

Question 4 continues on next page...

Question 4 continued…

iii. The implementation proposed in the previous question (Q4.c.ii) has an issue. That is if the main server crashes, the client could receive an incorrect answer.

Illustrate this problem in a time-sequence diagram. (*3 marks*)
How would you fix this problem? Present your solution in pseudo code. (*3 marks*)

[6 marks]

4.d Alice has developed a large-scale auctioning system with n replicas.

Consider the following scenario: a client sends a request to each of the n replicas and then waits for the servers to send back the result. Ideally, the client should receive n matching results. However, there could be up to *5* faulty replicas which *could give incorrect results or are not responding*.

i. Assume Alice uses *synchronous* group communications for client-server interactions and all server responses can be received in a short period of time.

To guarantee that the client always gets the correct answer, what is the minimum number of replicas that Alice needs?

[2 marks]

ii. Now assume Alice uses *asynchronous* group communication for client-server interactions. This means that the server responses could *be delayed for an indefinite amount of time*. In other words, a non-responding server will not be noticed by the client.

What is the minimum number of replicas that Alice needs, in order to ensure the client can always pick the correct answer in reasonable time?

[3 marks]

Total 25 marks

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